Single Oral Practice 1

Module 5 - Single Oral Administration · Daily practice sheet · No date

Name _____ Date _____

Procainamide is used for the treatment of ventricular tachyarrhythmia. It is administered intravenously, intramuscularly, and orally (Canada), and its therapeutic range is 4 to 8 mcg/mL. When a 750 mg dose is administered intravenously to a normal healthy subject: the elimination half-life is 3 hr; the apparent volume of distribution is 140 L or 2 L/kg; 65% is excreted in the urine; 35% is metabolized. When a 250 mg tablet is administered orally to a normal healthy subject: the absorption rate constant is 2.8 hr^{-1}; fraction of dose absorbed is 85.54%. Determine the following from the data provided:

Part 1 Total body clearance (and renal and metabolic clearance). What is the total body clearance?

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Part 2 Time to peak following the oral administration of a 250 mg dose.

Part 2 Maximum plasma concentration following the oral administration of a 250 mg dose.

Part 2 Maximum plasma concentration following the oral administration of a 500 mg dose.

Part 3 Whether the 250 mg tablet will provide a maximum plasma concentration sufficient to control the arrhythmia.

A. Yes: 1.22 mg/L is above 1 mcg/mL
B. No: its peak, 1.22 mg/L, is below the 4 mcg/mL lower limit
C. It cannot be judged without the AUC
D. Yes, after unit conversion 1.22 mg/L is 1220 mcg/mL

Part 4 If the 250 mg tablet is insufficient to control the arrhythmia, how many tablets (250 mg strength) will be required to control the arrhythmia?

A. 2 tablets
B. 4 to 6 tablets
C. 4 to 7 tablets
D. 3 to 6 tablets

Part 5 In a 70 kg patient with renal impairment, the elimination half-life is reported to be 14 hours. Assuming no change in apparent volume of distribution, absorption rate constant, and fraction absorbed systemically, determine the total body, renal, and metabolic clearances following the administration of a 250 tablet to this subject. What is the total body clearance?

Part 5 In a 70 kg patient with renal impairment, the elimination half-life is reported to be 14 hours. Assuming no change in apparent volume of distribution, absorption rate constant, and fraction absorbed systemically, determine the total body, renal, and metabolic clearances following the administration of a 250 tablet to this subject. What is the renal clearance?

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Part 6 Determine t_{max} and C_{max} for this patient with renal impairment (elimination half-life 14 hours) following a 250 mg oral dose. What is t_{max}?

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Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Part 1 Her key

Answer: 32.34 L/hr

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number*
D_0 IV	750 mg	*not needed: Cl_T depends on V_D and k alone*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	140 L or 2 L/kg	140 L multiplies k
f_e	65%	*not needed: the total is asked*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	*not needed*
F	85.54%	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = V_D \times k = (140\ L)(\frac{0.693}{3\ hr}) = 32.34\ L/hr$$

Total clearance $Cl_T = V_D \times k$, the volume of distribution times the elimination rate constant: each hour the fraction k of the drug in the 140 L is cleared. $k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$ from the IV half-life, and (140 L)(0.231 hr^{-1}) = 32.34 L/hr, litres times reciprocal hours.

> **DOES THE NUMBER MAKE SENSE?**
> A 3 hr half-life clears a little under a quarter of the 140 L each hour; a quarter of 140 is 35, so a Cl_T a little under 35, 32.34 L/hr, fits.

Part 1 Her key

Answer: 21.02 L/hr

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number*
D_0 IV	750 mg	*not needed: Cl_T depends on V_D and k alone*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	140 L or 2 L/kg	with k gave $Cl_T = 140 \times 0.231 = 32.34$ L/hr in part 1
f_e	65%	$f_e = 0.65$ multiplies Cl_T
metabolized	35%	*not needed: the renal part is asked*
k_a	2.8 hr^{-1}	*not needed*

F	85.54%	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = f_e \times Cl_T = (32.34\ L/hr)(0.65) = 21.02\ L/hr$

The stem says 65% is excreted in the urine unchanged, so $f_e = 0.65$ and the renal clearance is 0.65 of the total.

> **DOES THE NUMBER MAKE SENSE?**
> Sixty-five percent of 32.34 L/hr is about two thirds of it, near 21, so 21.02 L/hr; Cl_R must be below the total, 32.34 L/hr.

Part 1 Her key

Answer: 11.32 L/hr

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number*
D_0 IV	750 mg	*not needed: Cl_T depends on V_D and k alone*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	140 L or 2 L/kg	with k gave $Cl_T = 140 \times 0.231 = 32.34$ L/hr in part 1
f_e	65%	*not needed directly: $1 - f_e = 0.35$ is the metabolic share*
metabolized	35%	0.35 multiplies Cl_T
k_a	2.8 hr^{-1}	*not needed*
F	85.54%	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_H = (32.34\ L/hr)(0.35) = 11.32\ L/hr$

The stem says 35% of the dose is metabolized, so the metabolic (hepatic) clearance is 0.35 of the total clearance.

> **DOES THE NUMBER MAKE SENSE?**
> Thirty-five percent of 32.34 L/hr is about a third, near 11, so 11.32 L/hr; the metabolic part is below the renal part because 35% is less than 65%, and below the total.

Part 2 Her key

Answer: 0.97 hr

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number*

D_0 IV	750 mg	*not needed: t_{max} depends on k_a and k alone*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	140 L or 2 L/kg	*not needed*
f_e	65%	*not needed: the total is asked*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	in $\ln(\frac{k_a}{k})$ on top and in $k_a - k$ below
F	85.54%	*not needed*
D_0 oral	250 mg	*not needed: the dose does not move the peak time*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$$

k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$, the fraction of the drug eliminated per hour. t_{max}, the time of the peak, is when the absorption rate equals the elimination rate, so it depends on k and on k_a, the absorption rate constant, 2.8 hr^{-1}.

ALGEBRA

$$t_{max} = \frac{\ln(2.8/0.231)}{(2.8 - 0.231)\ hr^{-1}} = 0.97\ hr$$

$\frac{k_a}{k}$ is a ratio of two rate constants, a pure number, and dividing its log by $(k_a - k)$ in hr^{-1} gives hours.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption (k_a = 2.8 hr^{-1}) is about 12 times faster than elimination (k = 0.231 hr^{-1}), so the peak comes early, well within the first half-life of 3 hr; 0.97 hr fits.

Part 2 Her key

Answer: 1.22 mg/L

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number; the peak is compared with it afterwards*
D_0 IV	750 mg	*not needed: the oral dose is asked about*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	140 L or 2 L/kg	140 L in the denominator
f_e	65%	*not needed: the total is asked*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	in the coefficient and in $e^{-k_a t_{max}}$

F	85.54%	F = 0.8554 in the numerator
D_0 oral	250 mg	the dose in the numerator

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max} = \frac{F\ k_a\ D_0}{V_D(k_a - k)}\ (e^{-k\ t_{max}} - e^{-k_a\ t_{max}})$$

The single oral dose equation gives C_p at any time as $\frac{F\ k_a\ D_0}{V_D(k_a - k)}\ (e^{-kt} - e^{-k_a\ t})$; F = 0.8554 is the fraction absorbed, k_a = 2.8 hr^{-1} the absorption rate constant, k = 0.231 hr^{-1} the elimination rate constant. C_{max} is that equation at t = t_{max} = 0.97 hr.

ALGEBRA

$$C_{max} = \frac{(0.8554)(250\ mg)(2.8\ hr^{-1})}{140\ L\ (2.8 - 0.231)\ hr^{-1}}\ (e^{-(0.97)(0.231)} - e^{-(0.97)(2.8)})$$

Reciprocal hours cancel between k_a and (k_a − k), and mg ÷ L leaves mg/L, so the peak comes out as a concentration.

ROUND

$$C_{max} = 1.22\ mg/L$$

The peak is reported to two decimal places in mg/L, which is the same as mcg/mL for comparison with the range.

> **DOES THE NUMBER MAKE SENSE?**
> All of the absorbed dose, 0.8554 × 250 mg, in 140 L would give between 1 and 2 mg/L; the peak lies below that ceiling because elimination runs during absorption, so 1.22 mg/L fits and is below the 4 mcg/mL floor of the range.

Part 2 Her key

Answer: 2.44 mg/L

WHAT THE STEM GIVES

range	4 to 8 mcg/mL	*not needed for the number*
D_0 IV	750 mg	*not needed*
t½	3 hr	gives k = $\frac{0.693}{3}$ = 0.231 hr^{-1}
V_D	140 L or 2 L/kg	140 L in the denominator
f_e	65%	*not needed: the total is asked*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	in the coefficient and in $e^{-k_a\ t_{max}}$
F	85.54%	F = 0.8554 in the numerator
D_0 oral	500 mg	the dose in the numerator, twice the 250 mg tablet

THE WORKING, ONE LINE AT A TIME

SETUP

Same equation, D_0 = 500 mg; t_{max} is still 0.97 hr

The same oral equation with D_0 = 500 mg in the coefficient $\frac{F\,k_a\,D_0}{V_D(k_a - k)}$. $t_{max} = \frac{\ln(k_a/k)}{k_a - k}$ holds k_a = 2.8 hr^{-1} and k = 0.231 hr^{-1} but no dose, so the 500 mg dose peaks at the same 0.97 hr; only the coefficient changes.

ALGEBRA

$$C_{max} = \frac{(0.8554)(500\text{ mg})(2.8\text{ hr}^{-1})}{140\text{ L}\,(2.8 - 0.231)\text{ hr}^{-1}}\,(e^{-(0.97)(0.231)} - e^{-(0.97)(2.8)}) = 2.44\text{ mg/L}$$

With first-order kinetics the peak is proportional to the dose, so twice the dose gives twice the peak, 2.44 mg/L.

DOES THE NUMBER MAKE SENSE?

Twice the 250 mg dose gives twice the part 2 peak at the same t_{max}, 0.97 hr, so 2.44 mg/L; the whole absorbed dose in 140 L would be about 3 mg/L, a ceiling the peak stays under.

Part 3 Her key

Answer: B. No: its peak, 1.22 mg/L, is below the 4 mcg/mL lower limit

WHAT THE STEM GIVES

window	4 to 8 mcg/mL	the peak is compared with the lower limit, 4 mcg/mL
IV dose	750 mg	*not needed: the tablet is asked about*
t½	3 hr	gives k = $\frac{0.693}{3}$ = 0.231 hr^{-1}, inside t_{max} and C_{max}
V_D	140 L or 2 L/kg	140 L in the denominator of C_{max}
f_e	65%	*not needed: the split of elimination does not move the peak*
metabolized	35%	*not needed*
D_0	250 mg tablet	the dose in the numerator of C_{max}
k_a	2.8 hr^{-1}	in t_{max} and in the coefficient of C_{max}
F	85.54%	F = 0.8554 in the numerator of C_{max}

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{3\text{ hr}} = 0.231\text{ hr}^{-1};\ t_{max} = \frac{\ln 12.12}{(2.8 - 0.231)\text{ hr}^{-1}} = \frac{2.495}{2.569\text{ hr}^{-1}} = 0.97\text{ hr}$$

The peak of the 250 mg tablet, from part 2, is compared with the range, so it comes first: k from the IV half-life, then t_{max}; $\frac{2.8}{0.231}$ = 12.12 and ln 12.12 = 2.495.

ALGEBRA

$$C_{max} = \frac{(0.8554)(250\ \text{mg})(2.8\ \text{hr}^{-1})}{(140\ \text{L})(2.569\ \text{hr}^{-1})}\ (e^{-(0.231)(0.97)} - e^{-(2.8)(0.97)}) = (1.665\ \text{mg/L})(0.799 - 0.066) = 1.22\ \text{mg/L}$$

The coefficient is (0.8554)(250)(2.8) = 598.8 over (140)(2.569) = 359.7, which is 1.665 mg/L; the bracket is 0.799 − 0.066 = 0.733, so one tablet peaks at 1.22 mg/L.

UNIT

$$C_{max} = 1.22\ \text{mg/L} = 1.22 \times \frac{1000\ \text{mcg}}{1000\ \text{mL}} = 1.22\ \text{mcg/mL}$$

The range is in mcg/mL and the peak in mg/L. The factor of 1000 cancels, so 1 mg/L is 1 mcg/mL and the peak is 1.22 mcg/mL.

SETUP

Compare: C_{max} = 1.22 mcg/mL against the lower limit of the range, 4 mcg/mL; 1.22 is below 4

The peak has to reach at least 4 mcg/mL for the tablet to control the arrhythmia at any time. A peak of 1.22 mcg/mL never enters the range, so one 250 mg tablet is not sufficient.

> **DOES THE NUMBER MAKE SENSE?**
> mg/L and mcg/mL are the same size, so no factor of 1000 enters, and the whole absorbed dose, 0.8554 × 250 mg in 140 L, is under 2 mg/L; a peak above 4 mcg/mL from one tablet is impossible, and 1.22 fits below it.

1 mg is 1000 mcg and 1 L is 1000 mL, so the thousands cancel and the 1.22 mg/L peak is 1.22 mcg/mL. The peak is the highest the level gets after one tablet, and it never reaches the 4 to 8 mcg/mL range.

Part 4 Her key

Answer: B. 4 to 6 tablets

WHAT THE STEM GIVES

window	4 to 8 mcg/mL	both limits divided by the 1.22 mg/L one tablet gives
IV dose	750 mg	*not needed: the tablet is asked about*
t½	3 hr	gives $k = \frac{0.693}{3} = 0.231\ \text{hr}^{-1}$, inside t_{max} and C_{max}
V_D	140 L or 2 L/kg	140 L in the denominator of C_{max}
f_e	65%	*not needed: the split of elimination does not move the peak*
metabolized	35%	*not needed*
D_0	250 mg tablet	one tablet peaks at 1.22 mg/L; n tablets at n × 1.22
k_a	2.8 hr^{-1}	in t_{max} and in the coefficient of C_{max}
F	85.54%	F = 0.8554 in the numerator of C_{max}

THE WORKING, ONE LINE AT A TIME

SETUP

One tablet (part 2): $k = \frac{0.693}{3\ \text{hr}} = 0.231\ \text{hr}^{-1}$; $t_{max} = 0.97$ hr; $C_{max} = \frac{(0.8554)(250\ \text{mg})(2.8\ \text{hr}^{-1})}{(140\ \text{L})(2.569\ \text{hr}^{-1})}(e^{-(0.231)(0.97)} - e^{-(2.8)(0.97)}) = (1.665\ \text{mg/L})(0.799 - 0.066) = 1.22$ mg/L

Each 250 mg tablet adds the same 1.22 mg/L to the peak, because C_{max} is proportional to the dose. The number of tablets is a limit of the range over the peak of one tablet.

ALGEBRA

Lower limit: $\frac{4\ \text{mcg/mL}}{1.22\ \text{mcg/mL per tablet}} = 3.28$ tablets; $3 \times 1.22 = 3.66$ mg/L is below 4, $4 \times 1.22 = 4.88$ mg/L is inside

The minimum is the lower limit over the peak per tablet, 3.28. Three tablets leave the peak at 3.66 mcg/mL, below the range, so the count rounds up to 4 tablets, peaking at 4.88 mcg/mL.

ALGEBRA

Upper limit: $\frac{8\ \text{mcg/mL}}{1.22\ \text{mcg/mL per tablet}} = 6.56$ tablets; $6 \times 1.22 = 7.32$ mg/L is inside, $7 \times 1.22 = 8.54$ mg/L is above 8

The maximum is the upper limit over the peak per tablet, 6.56. Seven tablets push the peak to 8.54 mcg/mL, over the range, so the count rounds down to 6 tablets, peaking at 7.32 mcg/mL.

ROUND

3.28 rounds up to 4 tablets; 6.56 rounds down to 6 tablets: 4 to 6 tablets, peaks of 4.88 to 7.32 mcg/mL, inside 4 to 8

The minimum must clear the floor, so it rounds up; the maximum must stay under the ceiling, so it rounds down. Four to six tablets keep the peak inside 4 to 8 mcg/mL.

> **DOES THE NUMBER MAKE SENSE?**
> Each tablet peaks at 1.22 mcg/mL, so the count is a limit over 1.22; 3 tablets, 3.66, miss the floor of 4, and 7 tablets, 8.54, pass the ceiling of 8, so 4 to 6. Rounding both counts the same way puts one end outside the range.

C_{max} is proportional to the dose, so each tablet adds 1.22 mg/L to the peak. $\frac{4}{1.22} = 3.28$, so 4 tablets reach the minimum; $\frac{8}{1.22} = 6.56$, so 6 tablets stay under the maximum.

Her key writes 3.27 and 6.55 ($\frac{4}{1.22}$ and $\frac{8}{1.22}$ cut off at two decimals); the tablet counts are the same.

Part 5 Her key

Answer: 6.93 L/hr

WHAT THE STEM GIVES

weight	70 kg	*not needed: V_D is given as 140 L*

t½ impaired	14 hours	gives k = $\frac{0.693}{14}$ = 0.0495 hr⁻¹
V_D	140 L or 2 L/kg	140 L multiplies k
t½	3 hr	*not needed: replaced by 14 hours*
f_e	65%	*not needed: the total is asked*
metabolized	35%	*not needed*
k_a	2.8 hr⁻¹	*not needed*
F	85.54%	*not needed*
D_0 oral	250 tablet	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = (140\ L)(\frac{0.693}{14\ hr}) = 6.93\ L/hr$$

Total clearance $Cl_T = V_D \times k$, with $k = \frac{0.693}{t½}$. V_D, the volume of distribution, is unchanged at 140 L, but the half-life rose from 3 to 14 hr, so k falls from $\frac{0.693}{3\ hr}$ = 0.231 hr⁻¹ to $\frac{0.693}{14\ hr}$ = 0.0495 hr⁻¹; (140 L)(0.0495 hr⁻¹) = 6.93 L/hr.

> **DOES THE NUMBER MAKE SENSE?**
> The half-life rose from 3 to 14 hours, nearly five times, so clearance falls to about a fifth of part 1; k = $\frac{0.693}{14}$ hr⁻¹ times 140 L is 0.693 × 10 = 6.93 L/hr.

Part 5 Her key

Answer: 4.5 L/hr

WHAT THE STEM GIVES

weight	70 kg	*not needed: V_D is given as 140 L*
t½ impaired	14 hours	gives k = $\frac{0.693}{14}$ = 0.0495 hr⁻¹
V_D	140 L or 2 L/kg	with k gave Cl_T = 140 × 0.0495 = 6.93 L/hr in part 5
t½	3 hr	*not needed: replaced by 14 hours*
f_e	65%	f_e = 0.65 multiplies the new Cl_T
metabolized	35%	*not needed: the renal part is asked*
k_a	2.8 hr⁻¹	*not needed*
F	85.54%	*not needed*
D_0 oral	250 tablet	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_R = (6.93\ L/hr)(0.65) = 4.5\ L/hr$

The renal and metabolic shares are assumed unchanged at 65% and 35%, so Cl_R is 0.65 of the new, lower total clearance.

> **DOES THE NUMBER MAKE SENSE?**
> About two thirds of 6.93 L/hr is 4.5 L/hr, the renal share; Cl_R must be below the total of 6.93 L/hr.

Part 5 Her key

Answer: 2.43 L/hr

WHAT THE STEM GIVES

weight	70 kg	*not needed: V_D is given as 140 L*
t½ impaired	14 hours	gives $k = \frac{0.693}{14} = 0.0495\ hr^{-1}$
V_D	140 L or 2 L/kg	with k gave $Cl_T = 140 \times 0.0495 = 6.93$ L/hr in part 5
t½	3 hr	*not needed: replaced by 14 hours*
f_e	65%	*not needed directly: $1 - f_e = 0.35$ is the metabolic share*
metabolized	35%	0.35 multiplies the new Cl_T
k_a	2.8 hr^{-1}	*not needed*
F	85.54%	*not needed*
D_0 oral	250 tablet	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl_H = (6.93\ L/hr)(0.35) = 2.43\ L/hr$

The 35% metabolized share is unchanged by renal impairment in this problem, so Cl_H stays 0.35 of the new, lower total.

> **DOES THE NUMBER MAKE SENSE?**
> About a third of 6.93 L/hr is 2.43 L/hr, the metabolic share; it is below the renal share because 35% is less than 65%, and below the total of 6.93 L/hr.

Part 6 Her key

Answer: 1.467 hr

WHAT THE STEM GIVES

t½ impaired	14 hours	gives $k = \frac{0.693}{14} = 0.0495\ hr^{-1}$
t½	3 hr	*not needed: replaced by 14 hours*
V_D	140 L or 2 L/kg	*not needed: t_{max} depends on k_a and k alone*

f_e	65%	*not needed*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	unchanged; in $\ln(\frac{k_a}{k})$ on top and in $k_a - k$ below
F	85.54%	*not needed*
D_0 oral	250 mg	*not needed: the dose does not move the peak time*
range	4 to 8 mcg/mL	*not needed for the number*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{14\ hr} = 0.0495\ hr^{-1}$$

The 14 hr half-life gives a smaller elimination rate constant, 0.0495 hr^{-1}, in place of the 0.231 hr^{-1} for a healthy subject.

ALGEBRA

$$t_{max} = \frac{\ln(2.8/0.0495)}{(2.8 - 0.0495)\ hr^{-1}} = 1.467\ hr$$

$\frac{2.8}{0.0495} = 56.6$, a pure ratio of rate constants, and $\ln 56.6 = 4.036$; dividing by $(2.8 - 0.0495)$ $hr^{-1} = 2.7505\ hr^{-1}$ gives $\frac{4.036}{2.7505\ hr^{-1}} = 1.467$ hr. The peak comes later than with k = 0.231 hr^{-1} because less drug is eliminated while absorption continues.

> **DOES THE NUMBER MAKE SENSE?**
> Elimination is slower ($k = 0.0495\ hr^{-1}$) while $k_a = 2.8\ hr^{-1}$ is unchanged, so the peak comes later than in the healthy subject but still within the first 2 hours; 1.467 hr fits.

Part 6 Her key

Answer: 1.42 mg/L

WHAT THE STEM GIVES

t½ impaired	14 hours	gives $k = \frac{0.693}{14} = 0.0495\ hr^{-1}$
t½	3 hr	*not needed: replaced by 14 hours*
V_D	140 L or 2 L/kg	140 L in the denominator
f_e	65%	*not needed*
metabolized	35%	*not needed*
k_a	2.8 hr^{-1}	unchanged; in the coefficient and in $e^{-k_a t}$
F	85.54%	F = 0.8554 in the numerator
D_0 oral	250 mg	the dose in the numerator
range	4 to 8 mcg/mL	*not needed for the number*

THE WORKING, ONE LINE AT A TIME

ALGEBRA

$$C_{max} = \frac{(0.8554)(250\ \text{mg})(2.8\ \text{hr}^{-1})}{140\ \text{L}\ (2.8 - 0.0495)\ \text{hr}^{-1}}\ (e^{-(1.5)(0.0495)} - e^{-(1.5)(2.8)}) = 1.42\ \text{mg/L}$$

With the smaller k the peak is higher than the 1.22 mg/L in a healthy subject, because less drug is eliminated while absorption goes on.

DOES THE NUMBER MAKE SENSE?

Slower elimination leaves more drug at the peak, so C_{max} is above the healthy value but still below the ceiling of the whole absorbed dose in 140 L, 0.8554 × 250 mg over 140 L, a little over 1.5 mg/L; 1.42 mg/L fits.

Her key computes t_{max} = 1.467 hr and uses t = 1.5 hr here; C_{max} is 1.42 mg/L either way.
